

Stocks Are Not Flows:

An Identification Audit of Investor Pressure and Tenure Change in England

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Abstract

Can a rise in wealthy households' demand for housing explain falling owner occupation in England? The qualitative mechanism is plausible, but neither its direction nor its historical contribution is identified by tenure stocks, house prices, rents, aggregate mortgage rates, and net housing supply alone. We turn an initial structural project into a prospective identification audit. First, a source-level novelty review shows that the broad inequality–housing mechanism, endogenous rent–own–landlord choice, and linked owner-to-landlord flow measures all have close antecedents. Second, we separate three objects that are often conflated: tenure stocks, gross property-use transitions, and mechanism-specific resource flows. Third, a prospectively frozen internal fail-fast design evaluates a 57-moment, seven-mechanism response system against four deliberately adversarial omitted mechanisms. The maintained system is locally full rank at all 64 design points, yet three adversarial profiles exclude zero and required observation maps are absent. The authorized decision is therefore to stop unique attribution. A separate deterministic linear-algebra certificate sharpens the diagnosis: after appending the rival responses, the focal top-resource response lies in their span in both public and scaled units, so no linear discriminator exists. We then provide a general sharp-set formulation under bounded nuisance responses and state the measurement condition that new data must satisfy to break the equivalence. A hash-pinned public-data layer reproduces seven official series and 43 validation checks, including an 11.95 percentage-point fall in owner occupation among households whose reference person is aged 25–44 between 2008–09 and 2018–19. These accounting facts motivate the question but do not identify its cause. The paper's contribution is a reproducible framework for publishing a scientifically useful structural non-identification result and a concrete minimum-data agenda.

Keywords: housing tenure; private renting; wealth inequality; structural identification; partial identification; reproducibility.

JEL codes: C18, C52, R21, R31.

1 Introduction

The decline of owner occupation among younger and middle-aged households in England has invited a compelling narrative. If resources become concentrated at the top, wealthy households save a larger share, purchase existing housing, and let it to households priced out of ownership. The same transaction may then raise the private-rental stock, lower owner occupation, raise prices relative to rents, and reinforce measured wealth concentration. This narrative links Gary Stevenson’s fixed-asset models to a concrete tenure margin (Stevenson, 2019, esp. pp. 61–62). It is also easy to over-identify.

The empirical difficulty is not that the aggregate trends are invisible. It is that the objects needed to identify the mechanism differ from the objects in standard public time series. A tenure share is a stock. An owner-occupied home sold to a landlord is a gross transition. A wealthy purchaser’s incremental non-housing resources are a mechanism-specific forcing variable. A fall in the first does not reveal the second, and observing the second does not reveal the third. Prices create a further simultaneity: housing revaluation raises measured property wealth, so a contemporaneous wealth share can move because the proposed outcome moved.

This paper asks a deliberately prior question: *what would have to be observed, and what restrictions would have to hold, before the historical contribution of top-resource pressure could be distinguished from middle-resource loss, credit, rates, supply, taxation, demographics, rental segmentation, institutional entry, and social-comparison demand?* We answer it with an auditable sequence rather than a single fitted model.

The sequence has four parts. First, we conduct a source-level novelty gate. The broad sign pattern is occupied: inequality can lower ownership and raise price–rent ratios in equilibrium (Head et al., 2023); a poor-to-rich resource shift can induce wealthy portfolio reallocation into housing, higher prices, and lower rent–price returns (Zhang, 2015); and top-income growth can instead raise non-rich housing demand through social comparisons (Drechsel-Grau and Greimel, 2026). Close quantitative models already include endogenous tenure and household or institutional landlords (Moktan, 2025; Albuquerque et al., 2025; Cima and Kopecky, 2026; Castellanos et al., 2024). Moreover, linked Land Registry and rental-listing data have already been used to classify investor purchases and owner-to-landlord flows (Bracke, 2021; Bank of England, 2023). We therefore make no priority claim for the mechanism, the tenure margin, or the flow concept.

Second, we freeze an ontology and a 57-moment observation schedule before fitting the expensive model. The focal primitive is a latent aggregate flow of top non-housing resources, not observed saving, a wealth share, or a housing capital gain. Direct middle-resource loss and resource-neutral redistribution are separate experiments. Every synthetic shock has an explicit closure receipt.

Third, we run a fail-fast identification gate. The maintained seven-column response matrix is full rank at 64 design points, but the more demanding specificity tests fail. Three of four deliberately constructed rival data- generating processes produce a nonzero focal profile, and three economically essential stock-to-observation maps remain unimplemented. The gate stops the planned structural estimation rather than allowing fit to be mistaken for identification. A subsequent span calculation shows that the focal response lies in the admitted nuisance span. This is a deterministic statement

about the synthetic design, not a population false-positive rate or an empirical estimate.

Fourth, we retain what the evidence can support. A public accounting layer reproduces exact official-source paths, with immutable source hashes, strict spreadsheet schemas, deterministic figures, and a fresh offline replay. We also provide a general partial-identification formulation: conditional on a sourced response set and bounded nuisance contributions, the focal contribution is the projection of a linear feasibility set and can be computed sharply by two linear programs. No such empirical response set is presently justified for England, so we report the method but no historical interval.

The contribution is consequently an identification audit with a value-of- information result. It is positive in three senses. It separates commonly conflated estimands; it demonstrates that local full rank can coexist with failure against omitted mechanisms; and it converts a stopped structural model into a precise acquisition rule for future data. One well-oriented new measurement can break a focal–rival equivalence; another stock observation or rescaled version of the same aggregate series generally cannot.

2 From Stevenson’s thesis to a testable housing claim

2.1 What the 2019 thesis establishes

Stevenson’s MPhil thesis studies fixed and reproducible assets in stylised heterogeneous-agent economies in which asset holdings enter utility (Stevenson, 2019). Its central intuition is that a diminishing marginal propensity to consume can make a resource distribution relevant for the price of a fixed factor and its effective return. The thesis also recognizes its own load-bearing preference restriction. When rich households value land itself, the dynamic model can retain the inequality–price relationship. When they instead value fungible net worth, reproducible capital and land become direct substitutes and the steady-state land price collapses to the standard return-arbitrage relation $p = \rho/r$, independent of the distribution parameter in that exercise (Stevenson, 2019, pp. 48–49).

That negative robustness result is scientifically more informative than a mechanical extension of the preferred specification. The thesis does not estimate a tenure transition, observe a landlord purchase, model an empirically disciplined housing-supply margin, or identify a UK historical contribution. Its final chapter instead proposes a three-tier extension in which high asset prices may induce a middle class with small holdings to disavow to the rich (Stevenson, 2019, pp. 61–62). Our project began from that proposal, but the novelty and identification gates change what can responsibly be claimed.

2.2 Why a direct “three-class model” is not the residual novelty

Since 2019, and in some cases before it, adjacent literatures have professionalised nearly every broad component. Non-homothetic saving links inequality to lower rates and higher debt (Mian et al., 2021); falling rates can in turn affect the upper tail of wealth (Gomez and Gouin-Bonenfant, 2024). Housing models contain heterogeneous quality, search, rent–own choice, household landlords, institutional investors, credit constraints, and endogenous or fixed construction (Head et al., 2023;

Table 1: Novelty boundary after the source-level gate

Object	Closest occupied ground	Permitted residual claim
Inequality lowers owner-ship and raises price-rent	Equilibrium sign result in Head et al. (2023); supply-regulation and demand-side rivals	No new sign claim; test distinguishability of a separately defined forcing variable
Rich portfolio pressure raises house prices	Fixed-supply heterogeneous-agent mechanism in Zhang (2015)	No new mechanism claim; audit top-resource versus middle-resource and rival equivalence
Endogenous rent, own, and landlord choice	UK and European structural models including Moktan (2025), Albuquerque et al. (2025), and Cima and Kopecky (2026)	No architecture claim; state which observation equations identify which contribution
Owner-to-landlord property flows	Transaction-listing links in Bracke (2021) and Bank of England (2023)	No first-flow claim; distinguish flow classification from purchaser-resource identification
Historical England mechanism magnitude	No source located in the bounded review jointly observes the required resource and property-flow objects	Candidate only after an independently sourced response set; currently stopped

Notes: The review establishes collisions and a bounded residual; it is not proof of worldwide historical priority. Composite novelty by accumulation of familiar features is not asserted.

Greenwald and Guren, 2025; Cima and Kopecky, 2026). Zhang’s model is especially close to the portfolio-pressure story: wealthy households rebalance from bonds into housing after an inequality shift, while less-resourced non-owners enter later (Zhang, 2015). These antecedents validate the economic question but occupy the generic mechanism.

The residual object is narrower: a transparent test of whether an independently defined top non-housing resource flow can be distinguished from a direct loss of middle resources and from named rival mechanisms under England’s available observation map. This object remains useful even when the answer is no. It is not a claim to be the first study of housing flows: Bracke (2021) links transactions to subsequent rental listings, while the Bank of England classifies owner-to- landlord, landlord-to-owner, and inter-landlord flows using a longer listing window (Bank of England, 2023). Those sources also reveal why a public aggregate replication is insufficient: the classifications depend on proprietary listing links, are lagged and revisable, and do not observe purchaser non-housing resources.

3 Stocks, gross transitions, and resource contributions

3.1 The accounting non-equivalence

Let O_t , L_t , and S_t denote owner-occupied, privately let, and social or other housing-use stocks at date t . Abstracting temporarily from construction, demolition, vacancy, and reclassification, the change in privately let stock is

$$L_t - L_{t-1} = F_t^{O \rightarrow L} - F_t^{L \rightarrow O}. \quad (1)$$

The observed net change does not identify either gross flow. For every $k_t \geq 0$, the pair $(F_t^{O \rightarrow L} + k_t, F_t^{L \rightarrow O} + k_t)$ yields the same stock path. Adding landlord-to-landlord transactions changes turnover without changing the tenure stock. New construction, conversions, demolitions, vacancy, and household-dwelling denominator differences enlarge the equivalence class.

Even a correctly classified $O \rightarrow L$ purchase does not identify the proposed resource mechanism. The purchaser may be a leveraged household landlord, an existing cash landlord, a company, an institutional vehicle, or an owner temporarily letting a former residence. Its finance may reflect current saving, inheritance, capital gains, mortgage credit, retained rent, or asset substitution. The resource flow and the property-use transition are therefore distinct latent coordinates.

3.2 The focal primitive and closure

We write z_T for a latent aggregate increment to top non-housing resources and z_M for a middle-resource primitive. Neither is a contemporaneous wealth share. Three experiments have different economic meanings:

$$\begin{aligned} D1 : \quad & \Delta z_T > 0, \quad \Delta z_M = 0, \\ D2 : \quad & \Delta z_T = 0, \quad \Delta z_M < 0, \\ E0 : \quad & \Delta z_T = -\Delta z_M > 0. \end{aligned}$$

$D1$ and $D2$ require an explicit external or compositional closure and are diagnostic derivatives, not closed-economy redistributions. Their derivative sum represents $E0$ only locally, on the same differentiable equilibrium branch. This distinction prevents “pressure from above” from being defined mechanically as “impoverishment below.”

The measurement bridge also decomposes a non-housing wealth-stock change into active accumulation, asset returns, transfers, debt changes, valuation, and reclassification. Pre-period ranks are fixed locally. Property wealth and housing-secured debt are excluded from the focal primitive. These restrictions block the most direct reverse-causality channel but do not create an observed z_T series.

Table 2: The observation hierarchy and its claim ceiling

Layer	Example	Maximum interpretation
Stock	EHS owner and private-renter shares	Current weighted tenure composition; no gross transition
Net stock flow	Net additions divided by beginning dwelling stock	Change in the dwelling stock; no purchaser or tenure identity
Gross property-use flow	Transaction linked to pre/post rental listing	Classified $O \rightarrow L$, $L \rightarrow O$, or $L \rightarrow L$ transition under a stated window
Purchaser finance	Mortgage purpose, cash status, portfolio or legal form	Financing or buyer class; not top non-housing resources
Mechanism contribution	Response to an exogenous or bounded resource primitive	Structural or causal contribution only under the corresponding identifying assumptions

4 Public evidence and prospectively frozen scope

4.1 Source boundary

The public layer covers an England 2008–2019 core, while retaining each publisher’s stated geography. Seven official sources are pinned by exact URL, filename, media type, byte ceiling, and SHA-256: English Housing Survey (EHS) broad tenure, HM Land Registry HPI, separate ONS IPHRP and PIPR rent-index segments, dwelling stock, components of net housing supply, and Bank of England quoted mortgage rates (Ministry of Housing, Communities and Local Government, 2026a; HM Land Registry, 2026; Office for National Statistics, 2022, 2026; Ministry of Housing, Communities and Local Government, 2026b, 2025; Bank of England, 2026). The run rejects unexpected redirects, hash drift, schema drift, duplicate periods, missing registered support, and non-positive index or rate domains.

The open subset deliberately omits outputs that need a different assurance tier: EHS design covariance and fine tenure, Wealth and Assets Survey rank-by-tenure microdata, an audited Price Paid Data geography concordance, source-backed rival response sets, and a direct purchaser-resource link. Historical IPHRP and current PIPR are never level-spliced. The HPI/PIPR ratio is a normalized relative index, not a landlord yield. Quoted mortgage products are UK offers, not realized England borrowing costs.

4.2 Previously viewed outcomes and descriptive status

Feasibility work viewed the EHS endpoints and the 2015–2019 HPI/PIPR endpoint before the final implementation. These are therefore reproduced accounting facts, not untouched directional tests. No uncertainty interval is attached to the EHS age-band shares because the open table does not provide the required survey-design covariance. The analysis computes shares from unrounded weighted counts and checks their closure against a common denominator.

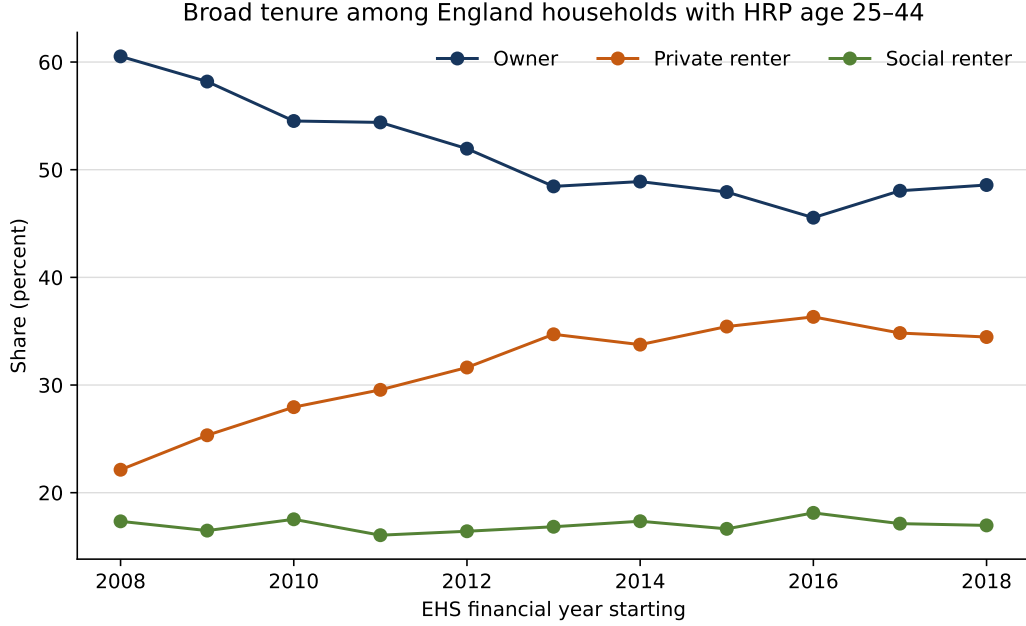
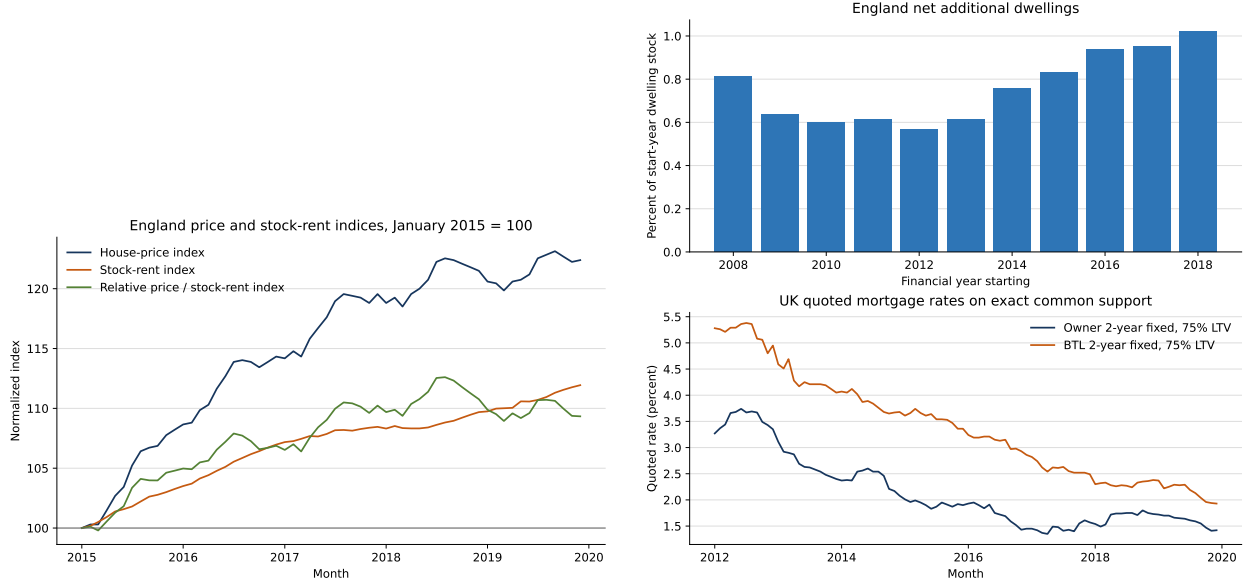


Figure 1: Broad tenure among English households whose household reference person is aged 25–44. Points are official weighted estimates reconstructed from unrounded counts. The open table does not support design-based confidence intervals for the combined age group.

Between 2008–09 and 2018–19, the owner share in this population falls from 60.5270% to 48.5783%, a change of -11.9487 percentage points. The private- renter share rises from 22.1325% to 34.4594%, or $+12.3269$ points; the social- renter residual changes by -0.3783 points. These arithmetic movements are close to mirror images, but they do not reveal the gross transactions, purchaser types, or resource origins that produced them.

The England HPI rises 29.5419% arithmetically between January 2008 and December 2019. Over their separate registered windows, IPHRP rises 12.6554% and PIPR rises 11.9390%; their levels are not spliced. On matched January 2015–December 2019 support, the normalized HPI-to-PIPR index rises 9.3346%. Net additions rise from 0.8119% to 1.0233% of the beginning-of-year dwelling stock between the first and last core years. On January 2012–December 2019 common support, the mean quoted BTL-minus-owner mortgage spread is 1.2573 percentage points. Each fact constrains a model, but none is a direct observation of z_T .

The Bank’s quoted-rate collection changed during 2019 from sampled end-month observations to whole-market monthly averages. We retain the published series as a descriptive credit-cost diagnostic and do not interpret the within-series break as an economic shock.



(a) House prices and the separate stock-rent segment. (b) Net additions and quoted mortgage rates.

Figure 2: Registered public accounting paths. The price-to-stock-rent series is normalized in January 2015 and is not a gross or net yield. Mortgage rates are quoted UK products; BTL support begins in January 2012.

5 A fail-fast identification design

5.1 Local response map

Let $g(\theta) \in \mathbb{R}^{57}$ map seven maintained mechanism amplitudes into the registered target vector, after an explicit public-unit observation adapter. At a design point θ_0 , the local response matrix is

$$J(\theta_0) = \left[\frac{\partial g}{\partial z_T}, \frac{\partial g}{\partial z_M}, \frac{\partial g}{\partial c}, \frac{\partial g}{\partial s}, \frac{\partial g}{\partial \tau_L}, \frac{\partial g}{\partial r}, \frac{\partial g}{\partial d} \right], \quad (2)$$

where the remaining columns denote credit, planning or supply, landlord tax, interest-rate, and demographic blocks. The coefficients are synthetic design inputs, not empirical elasticities. Percentage-level tenure and pounds-per-week rent units are preserved before a separately frozen diagonal scaling is applied.

Full column rank of J is necessary for local separation inside the maintained model but is not sufficient for mechanism specificity. An omitted response can project onto the focal column even when the maintained columns are linearly independent. We therefore require known-negative rank controls, deterministic adversarial data-generating processes, profile diagnostics, and executable maps between model stocks and aggregate published outcomes before expensive recovery experiments may run.

5.2 Prospective decision rule

The gate was frozen to return either `FAIL-IDENT-0` or `CONTINUE-TO-FULL-GATE`; it has no path to declare final identification. A controlling failure stops the 6,573 planned fits and 210,336 optimizer starts. This is a scientific stop rule, not merely a computational shortcut: once a cheap test demonstrates that unique attribution is unsupported, running a large model cannot repair the missing information.

Four deliberately adversarial rival responses represent social comparison, institutional entry, rental segmentation, and planning restriction. They are stress constructions, not a random sample of mechanisms. Accordingly, the share of those four cases producing a nonzero profile is never interpreted as a population false-positive rate.

6 Identification geometry and conditional sharp sets

Partition a local linear response as

$$y = f\beta + N\gamma, \quad (3)$$

where $f \in \mathbb{R}^m$ is the focal top-resource response, $N \in \mathbb{R}^{m \times k}$ contains maintained nuisance and admitted rival responses, β is the focal amplitude, and γ collects nuisance amplitudes. The following elementary result is the paper’s central diagnostic.

Proposition 1 (Linear discriminator). *There exists a vector $d \in \mathbb{R}^m$ satisfying $d^\top N = 0$ and $d^\top f = 1$ if and only if $f \notin \text{col}(N)$. When $r = (I - P_N)f \neq 0$, where P_N is the orthogonal projector onto $\text{col}(N)$, one such discriminator is $d = r/(r^\top r)$.*

Proof. If $f = Nc$ for some c , then every d with $d^\top N = 0$ also satisfies $d^\top f = d^\top Nc = 0$, contradicting $d^\top f = 1$. Conversely, r is orthogonal to $\text{col}(N)$ and $r^\top f = r^\top r > 0$, so the stated normalization has both required properties. $\square$

The numerical certificate normalizes each nonzero nuisance column, forms a truncated-SVD projector, and compares the residual of a unit-norm focal response with a prospectively fixed tolerance. This makes the span decision invariant to units without obscuring the original column norms. Repeated time ramps or rescaled copies of existing moments cannot create a discriminator because they do not add a new response direction.

Point identification is not the only coherent target. Suppose nuisance amplitudes have externally justified bounds $\ell \leq \gamma \leq u$, with optional bounds on β . The sharp conditional focal set is

$$\mathcal{B}(y) = \{\beta : \exists \gamma : y = f\beta + N\gamma, \ell \leq \gamma \leq u\}. \quad (4)$$

Because the feasible set is convex, its projection onto β is an interval. Its endpoints solve two linear programs. Infeasibility and unboundedness are substantive outcomes, not software errors. What gives such an interval empirical content is the provenance of f , N , y , and the boxes—not the optimizer. The current England application lacks independently source-backed response sets, so we do not calculate an empirical interval.

7 Results

The empirical programme was governed by a sequence of stopping rules rather than by an obligation to produce a structural estimate. The first rule tested whether the planned aggregate observation schedule could distinguish a latent top non-housing resource-flow amplitude, denoted z_T (machine label `z_top`), from six maintained blocks and four deliberately adversarial omitted mechanisms. That test returned `FAIL-IDENT-0`. The failure stopped the registered minimum executable structural model and any unique historical attribution. We therefore report three objects with different evidential status: the authoritative `IDENT-0` design diagnostic, a licence-safe reproduction of an open public-accounting subset, and a deterministic design-geometry certificate. None is a causal estimate of wealth-driven tenure change.

7.1 `IDENT-0`: local rank without mechanism specificity

The authoritative v2 run adapted all 57 registered targets to their declared public units and assigned a unique valid accounting receipt to each of 313 closure cases. The maintained seven-column response matrix passed the local rank front at all 64 design points and at each of three central-difference step sizes. Across those evaluations, the scaled condition number ranged from 62.3556 to 64.0830. The known-negative controls behaved as registered: exact top-middle collinearity reduced rank from seven to six, near collinearity raised the condition number above 3.418×10^8 , and an exact top mimic was detected.

These checks establish that the restricted maintained design is numerically non-degenerate; they do not establish specificity. In the adversarial part of the harness, three of four deliberately constructed omitted-mechanism cases induced a z_{top} profile that excluded zero when fit by the restricted seven-block model. This is a deterministic stress-harness result, not a population frequency or a false-positive rate. In none of the four cases was z_{top} co-largest in absolute value, and all 32 starts in each case converged to the same basin. The three missing observation maps—owner and landlord prices to aggregate HPI, dwelling stocks to household tenure, and lifecycle states to the age-25–44 population—were independent reasons to fail the gate. Table 3 records the controlling criteria.

Because the fail-fast front was decisive, the precomputed full-stage workload of 6,573 fits and 210,336 optimiser starts was not run. In particular, the 313-case recovery aggregation, noisy-fit coverage and classification checks, 64-case profile aggregation, rival-compensation tests, outer optimiser rule, structural mutations, and the P5 landlord-yield signature remain unexecuted. The run did not issue, and could not be read as issuing, a pass for unique attribution.

7.2 Gate A: reproduced public accounting, with the full gate held

After the identification stop, Gate A was restricted to deterministic accounting over seven official public sources. All seven downloaded objects matched their prospectively frozen SHA-256 digests, and all 43 schema, support, closure, and arithmetic checks passed. This produced nine derived tables and three figures in PDF and SVG. Its only authorised success label

was `OPEN-SUBSET-REPRODUCED+FULL-GATE-A-HELD`; the runner had no code path to authorise structural estimation, unique attribution, or causal language.

For private households in England whose household reference person was aged 25–44, the open English Housing Survey table gives an owner point estimate of 60.5270% in 2008–09 and 48.5783% in 2018–19, a change of -11.9487 percentage points (Table 4). The private-renter estimate moved from 22.1325% to 34.4594%, or $+12.3269$ percentage points; the social-renter estimate changed by -0.3783 percentage points. These shares were recomputed from unrounded weighted counts over a common denominator. The maximum discrepancy from published age-specific percentages was 1.99×10^{-13} percentage points, and the largest absolute three-tenure closure residual was 2.84×10^{-13} percentage points. The public table supplies neither the design covariance needed for joint inference nor the registered fine-tenure categories, so these are point estimates without confidence intervals.

The England all-property HPI rose from 63.3 in January 2008 to 82.0 in December 2019, an endpoint change of 29.5419% (or 25.8834 log points). The historical IPHRP stock-rent segment rose 12.6554% over 2008–2014, and the methodologically separate PIPR segment rose 11.9390% over 2015–2019. Those levels were not spliced. On their common 2015–2019 support, normalising both HPI and PIPR to January 2015 yields a price-to-stock-rent relative index that rose 9.3346%; its reciprocal fell 8.5377%. This relative index is not a gross or net rental yield.

Net additions increased from 0.8119% of the start-of-year dwelling stock in 2008–09 to 1.0233% in 2018–19. The denominator is all dwellings, including vacant dwellings, and is not interchangeable with the household denominator used for tenure. The maximum absolute residual between the published net-additions total and its components was 13 dwellings, below the pre-run quality-control tolerance of 50. Finally, on the 96-month common support from January 2012 to December 2019, the quoted two-year fixed 75% LTV buy-to-let minus owner-occupier mortgage-rate spread declined from 2.01 to 0.51 percentage points and averaged 1.2573 points. These are United Kingdom quoted product rates, not realised borrowing costs for English households. The publisher’s 2019 change from sampled end-month observations to whole-market monthly averages is a measurement break, so the series is retained only as a descriptive diagnostic rather than interpreted as a policy or market shock.

Full Gate A remains held. The open EHS table lacks design covariance and fine tenure; net non-housing wealth by rank and tenure requires Wealth and Assets Survey EUL microdata; the Price Paid Data turnover module requires an audited England/Wales concordance; and source-backed rival response sets remain absent. The FCA mortgage table and the English Private Landlord Survey are bounded extensions rather than inputs to this run. Most importantly, the unrestricted open core does not jointly label gross property-use transitions, purchaser landlord identity, and purchaser top non-housing resources. Existing proprietary and safeguarded listing linkages remain relevant prior evidence; their existence also precludes any claim that property-use flows have never been measured.

7.3 Partial identification: a design-geometry obstruction

The post-gate certificate asks whether the focal response column has a component outside the span of the other six maintained columns and the four admitted rival signatures. Within the maintained 57-by-7 matrix, rank is seven in both public units and the prospectively frozen synthetic diagonal scaling. After adding four rival columns, the 57-by-11 matrix has rank nine. Projection of the focal z_{top} column onto the ten nuisance and rival columns leaves relative residuals of 1.461×10^{-14} in public units and 2.323×10^{-14} after synthetic scaling, both far below the registered 2×10^{-10} effective tolerance (Table 5). No dual discriminator exists at that tolerance.

The certificate decision is **STOP-UNIQUE+GO-PARTIAL-ID-THEORY**. It reconciles two facts that would otherwise appear contradictory: the focal coefficient is locally recoverable inside the restricted maintained matrix, yet is not identified against the enlarged admitted span. This is a deterministic algebraic statement about the supplied columns. All maintained loadings, baselines, scales, and rival signatures are synthetic design inputs, and the four rivals are constructed stress cases rather than draws from any population. No response set in this certificate has been independently source-backed or transported to England. Consequently, the certificate supplies neither an England elasticity nor an empirical interval for the historical contribution of top resources.

The accompanying linear-programming layer can compute a sharp interval for a focal coefficient conditional on an observed response equation and declared box restrictions on nuisance coefficients. Promotion of such an interval to an empirical England bound would require independently sourced mechanism response sets, explicit geography/time/population/unit transport sets, and sensitivity to enlarging the rival set and relaxing its boxes. Those inputs do not yet exist, so no numerical England contribution bound is reported.

Table 3: Authoritative IDENT-0 v2 criteria.

Criterion	Status	Registered threshold or control	Realised value and interpretation
Public-unit observation schedule	Pass	57 targets adapted; no pending adapter	57 targets; H05 retained as an explicit non-modelled holdout.
Signed closure coverage	Pass	One unique, valid receipt for every case	313/313 cases, using the registered elementary, composite-open, or no-change taxonomy.
Maintained local identification	Pass	At least 61/64 rank-seven points; midpoint must pass	64/64 points passed at all three step sizes; scaled condition numbers 62.3556–64.0830.
Exact- and near-collinearity controls	Pass	Exact mutation must lose rank; near mutation must exceed 10^4	Exact ranks 6, 6, 6; near-collinear condition numbers approximately 3.418×10^8 .
Constructed-rival profile diagnostic	Fail	No more than 0.05 of the four registered profiles may exclude zero	3/4 profiles excluded zero. This proportion describes four constructed cases only.
Constructed-rival top-largest diagnostic	Pass	No more than 0.10 may make z_{top} co-largest	0/4 cases made z_{top} co-largest.
Constructed-rival optimiser stability	Pass	At least 30/32 starts per case in the same basin	32/32 for each of four cases.
Exact top-mimic detection control	Pass	Harness must expose attribution under identical labels	The control returned $\hat{z}_{\text{top}} = 0.65$ and profile $[0.375, 0.925]$.
Three observation bridges	Fail	Each bridge must be executable	All three were NOT_IMPLEMENTED: HPI aggregation, dwelling-to-household tenure, and age-25–44 lifecycle aggregation.

Notes: the machine report contains 12 criteria in total: eight pass and four fail. The final row consolidates three separately recorded failures. Synthetic response coefficients, terminal wealth loadings, and diagonal scales are recovery-design inputs, not empirical elasticities or sampling standard errors.

Table 4: Reproduced Gate A public-accounting facts.

Registered fact	Reproduced accounting result	Interpretation ceiling
AF01: owner tenure, HRP 25–44	60.5270% (2008–09) to 48.5783% (2018–19); –11.9487 percentage points	England point estimates; public design covariance unavailable.
AF02: private renting, HRP 25–44	22.1325% to 34.4594%; +12.3269 percentage points	Same household denominator and uncertainty restriction as AF01.
AF03: broad-tenure path	11 financial years, 2008–09 to 2018–19; owner/private/social only	Fine tenure and joint covariance remain EUL-blocked.
AF07: England HPI	63.3 (January 2008) to 82.0 (December 2019); +29.5419%	Unadjusted all-property index; not a landlord-specific price.
AF08: stock-rent segments	IPHRP +12.6554% (2008–2014); PIPR +11.9390% (2015–2019)	Separate method segments; no level splice.
AF09: normalised price/stock-rent index	+9.3346% over January 2015–December 2019; reciprocal –8.5377%	A relative index, not a rental yield.
AF11: net additions rate	0.8119% (2008–09) to 1.0233% (2018–19)	Flow divided by start-of-year all-dwelling stock, including vacant dwellings.
AF12: quoted mortgage-cost diagnostic	96 common months; BTL-minus-owner spread 2.01 to 0.51 points; mean 1.2573	UK quoted offers, not realised England borrower costs.

Notes: EHS endpoints and the 2015–2019 HPI/PIPR endpoint had been viewed during feasibility work. Their manifest-bound calculation is therefore a reproduction of previously observed accounting facts, not an untouched directional confirmation. Passing arithmetic checks authenticates the registered transformation; it does not validate a causal mechanism.

Table 5: Partial-identification design-geometry v1 certificate.

Basis	Maintained rank	Condition no.	Augmented rank	Relative residual
Public units	7/7	68.8814	9/11	1.46097×10^{-14}
Synthetic scaling	7/7	63.3405	9/11	2.32265×10^{-14}

Notes: columns are individually normalised for the span calculation. The maximum absolute difference between the analytic public-unit derivative and its central-difference control was 1.34432×10^{-9} . The augmented condition number is intentionally not reported because the augmented matrix is rank-deficient.

Table 6: Manifest- and report-bound run identities.

Run	Manifest SHA-256	Report SHA-256
IDENT-0 v2	a23d17ea471eb7ed67a96e34b5fb85f7 c50b34c7e231987498ed36963220ccda	049ec5f8c3e290d1c0273c8630e3c8d1 eeee8c2191427fcd9742803060bfca35
Gate A v2, online	3e1f92e436e3744d6702d8314c9f4c0f 5602898d5b996f9a60fa0010af291c90	97925770879ca8f9e356c067c9d9c6b4 b98d5d098113354485002bed8bcc2f08
Gate A v2, offline replay	f00bfcace904d7a3cb6fcb8746d1fb03 46b38e74de6d67224d868ccb0a03a57b	51aa01e7a0b2fe14aed73e3d3ff43b53 3637f1e43f91a0bc96598ef54f1cc530
Design geometry v1	7e76a2930297e0b13ad2cd6aa1a6a4a4 3cbe0a478da2405011a30fadba51b5d	421ce83b335bf610e2f7fa3cfb178b6a 2ebc7620098d05b9f5601d164cb1d7fc

Table 7: Registered source-object digests for Gate A v2.

Source object	SHA-256
EHS FA1201 open table	f095268b6330ec5e4733b0474f53d8b79d7a6932dc939e0b65196799 d54252ac
HM Land Registry HPI indices, May 2026	cacaef81b8f09c0b1c67c9a8a7c077134384a1be4dd1b1ddf26a9c52 656620a5
ONS historical IPHRP time series, version 25	bf87cfd293323df5c558a203635d69c16b9b1c48485d410afc19563c 016228f1
ONS PIPR workbook, 22 July 2026	d429e553ade454903b731f6c257609f14d0c2171822791e09c21a9d5 3d5e24a9
MHCLG dwelling stock Live Table 104	ecd02eb8e520e4fef7878bda35db002ca2360cb45112416db6619863 b211a1fa
MHCLG net supply Live Table 120	548101e535b7db4e0718a1ef3129e343169cb3982f1cd8e03c0f247e 93c5d219
Bank of England quoted-rate extract	d5c17cab1103c3396695f389c6384b745ce58e8f4f07787f83d82a6 2127f009

8 Reproducibility and assurance

8.1 Immutable run identities

Every reported run required a clean Git worktree and recorded the command, commit, configuration and source-code digests, Python platform, NumPy and SciPy versions, and the fixed provenance seed 20260801. IDENT-0 v2 ran at commit 1e04a181783105051c819af190b9546340bbb1fd; Gate A v2, its offline replay, and the partial-ID certificate ran at commit 836e3391355f84438af0d321f69e94281050c00d. The authoritative identities are:

The seven official Gate A source objects were also pinned before extraction (Table 7). A matching digest establishes byte identity with the registered source object; it does not establish survey validity, measurement accuracy, or a causal interpretation.

8.2 Online acquisition and offline replay

Gate A v2 was first run against the registered official HTTPS locations and then rerun using only the immutable raw directory from the online run. The seven raw files, all nine derived CSVs, all

six rendered figure files, and `TABLE_METADATA.json` were byte-identical across the two runs. The scientific content of `REPORT.json` was also identical after excluding its run-manifest digest. The complete report and directory digests differ by design: the online and offline manifests record different commands, and their source receipts record official-HTTPS retrieval versus offline-cache replay. This is therefore an exact replay of the pinned scientific payload, not a claim that provenance-bearing directories are byte-for-byte identical.

From the repository root, the manifest-recorded commands (with the local virtual-environment prefix shortened to `.venv/bin`) are:

```
.venv/bin/housing-ident0 -config configs/ident0.json
                        -output-dir results/ident0/20260801-ident0-failfast-v2
                        -exit-zero-on-scientific-fail

.venv/bin/housing-gatea -config configs/gatea_public.json
                        -output-dir results/gatea/20260801-open-subset-v2

.venv/bin/housing-gatea -config configs/gatea_public.json
                        -offline-source-dir results/gatea/20260801-open-subset-v2/raw
                        -output-dir results/gatea/20260801-open-subset-v2-replay

.venv/bin/housing-partialid -config configs/ident0.json
                        -ident0-report results/ident0/20260801-ident0-failfast-v2/REPORT.json
                        -output-dir results/partialid/20260801-design-geometry-v1
```

The IDENT-0 option `-exit-zero-on-scientific-fail` changes only the shell exit status used by the release workflow; it does not change the `FAIL-IDENT-0` decision written into the report. The Gate A downloader fails closed on host, media-type, byte-ceiling, schema, support, and digest drift. Scientific invariants use explicit exceptions rather than Python `assert`, and the focused test protocols include optimised `python -O` execution so that disabling assertions cannot erase those checks.

8.3 Assurance boundary and remaining work

The combined evidence supports three deliberately narrow conclusions. First, the prospectively frozen aggregate design does not support unique attribution of England’s tenure change to the focal resource mechanism once the registered adversarial rivals and missing observation bridges are taken seriously. Second, a bounded collection of public tenure, price, rent, housing-supply, and quoted-rate paths is reproducible from pinned official bytes. Third, within the synthetic design harness, the focal response lies in the enlarged nuisance and rival span at the registered numerical tolerance.

It does not support a causal top-wealth-to-tenure claim, a historical contribution percentage, a general-equilibrium result, an empirical England elasticity, or an England partial-identification interval. Hashes authenticate the bytes and declared replay boundary, not the correctness of publisher

methods or the truth of an economic mechanism. Deterministic replay is not independent replication. Likewise, agreement between an analytic derivative and a finite-difference control validates an implementation detail, not its empirical calibration.

An empirical promotion would require, at minimum, survey-design covariance and fine-tenure microdata; net financial plus business wealth by resource rank and tenure; an audited map between dwelling flows and household tenure; direct or defensibly bounded purchaser-resource information; and independently source-backed response sets for every admitted mechanism with explicit transport across geography, period, population, unit, and scale. Until those conditions are met, the appropriate endpoint is the present identification audit and public-accounting package, with **STOP-UNIQUE** and **STOP-CAUSE** retained.

9 Minimum-data and value-of-information result

The failed gate does more than request “better data.” It specifies the direction that a useful measurement must add. Let $C = \{c : Nc = f\}$ be the set of nuisance representations of the current focal response. Append one new measurement with focal response a_f and nuisance row $a_N^\top$. The extended columns are

$$f^+ = \begin{bmatrix} f \\ a_f \end{bmatrix}, \quad N^+ = \begin{bmatrix} N \\ a_N^\top \end{bmatrix}.$$

Proposition 2 (A measurement that breaks observational equivalence). *If $f \in \text{col}(N)$ in the current observation map, then the appended measurement identifies a new focal direction if and only if*

$$a_f \notin \{a_N^\top c : c \in C\}. \quad (5)$$

Equivalently, $f^+ \notin \text{col}(N^+)$. Thus one well-oriented scalar measurement can be sufficient, while arbitrarily many measurements satisfying the same nuisance representation cannot help.

Proof. $f^+ \in \text{col}(N^+)$ precisely when there exists c such that both $Nc = f$ and $a_N^\top c = a_f$. This is equivalent to a_f belonging to the set on the right-hand side of equation (5). Negating the condition and applying proposition 1 gives the result. $\square$

The synthetic augmented matrix has 11 columns and rank nine, but its two-dimensional algebraic rank deficit is not a claim that exactly two real-world variables are required. The orientation of a measurement matters more than its count. In housing terms:

1. another tenure-stock year, or the same stock expressed in a different denominator, is unlikely to break the equivalence;
2. a validated $O \rightarrow L$ transaction label reduces gross-flow ambiguity but still does not reveal the purchaser’s resource shock;
3. a top non-housing wealth measure without a property linkage remains vulnerable to asset returns, transfers, debt changes, and aggregation; and
4. the most valuable design jointly links pre-determined non-housing resources or a defensible resource shock to purchaser identity, property-use transition, financing, and a comparison group, with supply and credit responses measured on compatible support.

This joint object could arise from securely linked tax, mortgage, ownership, and rental-listing records, or from a narrowly valid policy or windfall design. Access alone is not identification: linkage error, anticipation, legal-form switching, reclassification, and equilibrium spillovers must enter the response set. The criterion in equation (5) provides a pre-analysis value-of-information test for candidate acquisitions.

10 Discussion

10.1 What the failure establishes

The results establish that the registered synthetic observation map does not support unique top-resource attribution. This conclusion is stronger than noting that a regression may be endogenous and weaker than proving that no conceivable England design can identify the mechanism. It is bound to the admitted response class. A new source or restriction can alter the column space; it must be justified without using the desired conclusion.

Local full rank and low residual loss inside a maintained model should therefore not be reported as identification when omitted economically plausible mechanisms have aligned responses. The known-negative controls in this project illustrate the difference: the code correctly rejects exact and near collinearity, yet the apparently healthy seven-column design remains non-specific after rivals are admitted. The problem is semantic coverage, not numerical optimization.

10.2 What the public facts do and do not show

The open data show a large reallocation of broad tenure among households aged 25–44, rising house prices, rising stock rents, changes in relative prices, a recovering net-supply rate, and a narrowing quoted BTL mortgage spread over the registered window. These are important constraints. They do not determine whether the tenure shift arose from rich household purchases, institutional entry, middle-resource loss, credit conditions, supply, demographics, or combinations. Nor can the near-offsetting owner and private-renter share changes be treated as gross owner-to-landlord transfers.

The linked-listing literature is informative but does not close the mechanism. Bracke (2021) defines investor acquisition through a rental listing within six months of sale. The later Bank measure uses a four-year window and prior/subsequent listing histories to classify three property flows (Bank of England, 2023). These are different estimands; both are sensitive to coverage, timing, and classification; online-listing volume is itself sensitive to platform coverage (Livingston et al., 2021). Neither directly identifies purchaser wealth rank or the origin of funds. A future paper should treat their definitions as measurement models, not ground-truth labels.

10.3 Relation to partial identification

Partial identification is not a rhetorical downgrade of point identification. It replaces a false singleton with a transparent set whose width reflects admitted uncertainty (Manski, 2003; Tamer, 2010). The sharp-set machinery in this paper is ready to accept source-backed mechanism response ranges. It deliberately refuses to transform the synthetic IDENT-0 loadings into an England interval. Informative bounds may eventually support a near-zero, signed, or sign-ambiguous conclusion; all are scientifically admissible.

10.4 Limitations

Four limitations define the paper’s claim ceiling. First, the public tenure path contains point estimates without the design covariance needed for combined-age inference. Second, the design responses and adversarial rivals are synthetic. They test a planned model, not England. Third, the novelty search is bounded and cannot prove global priority. Fourth, the analysis does not estimate a causal or structural historical contribution. These limitations are not buried robustness notes: they determine the title, abstract, and decision code.

11 Conclusion

The project began with an intuitively attractive extension of Stevenson’s thesis: wealth concentration raises fixed-asset demand, encourages middle-class dissaving, and reallocates owner-occupied homes to landlords. The novelty gate showed that the broad theoretical and empirical territory is already occupied. The identification gate then showed that the registered aggregate observation map cannot distinguish the focal top-resource response from admitted alternatives. The responsible outcome is not to force a structural estimate. It is to publish the boundary.

Three results survive that boundary. Official public data reproducibly document the motivating accounting paths. A discriminator theorem and manifest-bound certificate state exactly why unique attribution fails in the planned design. A conditional sharp-set framework and new-measurement criterion turn the failure into a concrete data-acquisition agenda. In this setting, “stocks are not flows” is not merely a warning about terminology. It is the identification result.

Declarations

Data and code availability. The replication package contains source identities and hashes, extraction and validation code, tests, machine-readable reports, deterministic figures, and a deterministic offline replay. Restricted, safeguarded, proprietary listing, and address-level data are not redistributed. Exact release identifiers are reported in the reproducibility appendix. The archival preprint and replication package are at doi:10.5281/zenodo.21740339; the public source repository is <https://github.com/ipitchford/stocks-are-not-flows>.

Internal protocol and outcome viewing. The public pilot was governed by a prospectively frozen internal protocol and three amendments. The EHS endpoints and the 2015–2019 HPI/PIPR endpoint had been viewed during feasibility work. The net-supply component tolerance was fixed during pre-run schema inspection. These disclosures preclude an untouched confirmatory interpretation.

Competing interests and funding. No funding or competing-interest declarations were supplied for this anonymous public discussion preprint. These declarations, affiliations, and accountable human authorship must be completed before journal submission.

Use of computational assistance. Automated research and coding tools assisted source discovery, implementation, testing, and manuscript drafting. Human authors remain responsible for source verification, the scientific claims, and the submitted text. Automated tools are not authors.

A Identification and stock–flow accounting results

A.1 Scope, novelty, and assurance boundary

This appendix states the mathematical results used to interpret the empirical design. The rank identities, projection arguments, linear-programming characterisation, and stock–flow inequalities below are standard consequences of finite-dimensional linear algebra, convex analysis, and accounting. We do not claim them as new theorems. The housing-specific contribution is instead the disciplined application of these results to a prospectively frozen observation map, the separation of stocks from gross flows, and the use of explicit response sets and provenance gates before attaching an empirical interpretation.

Nothing in this appendix supplies a numerical bound for England. In particular, a response column used in a design audit is not thereby established as a historical response, a causal impulse response, or a transportable elasticity. Any empirical interval requires independently sourced response sets, declared population and time transport assumptions, and a separate measurement audit. The propositions below describe what follows *conditional on* their stated matrices and restrictions.

A.2 Linear response setup

Let $y \in \mathbb{R}^m$ denote an observed vector of moments. The exact linear response model is

$$y = f\beta + N\gamma, \tag{6}$$

where $f \in \mathbb{R}^m$ is the response column for the focal mechanism, $\beta \in \mathbb{R}$ is its coefficient, $N \in \mathbb{R}^{m \times k}$ collects admitted nuisance and rival response columns, and $\gamma \in \mathbb{R}^k$ contains their coefficients. Zero or linearly dependent columns of N are allowed. Unless restrictions are introduced explicitly, γ is unrestricted.

The phrase *identified against the admitted span* is deliberate. It is a statement about the columns included in N , not a guarantee that all economically relevant mechanisms have been included. Enlarging N can remove identification; validating the membership of N is therefore a substantive part of the empirical design rather than a purely numerical step.

A.3 A common time path cannot create a new identifying direction

Suppose $a_j \in \mathbb{R}^q$ is the cross-moment loading for mechanism j , and all mechanisms are multiplied by the same time path $h \in \mathbb{R}^T$. With moments stacked by date, the corresponding design is

$$D = h \otimes A, \quad A = \begin{bmatrix} a_1 & \cdots & a_p \end{bmatrix} \in \mathbb{R}^{q \times p}. \tag{7}$$

Proposition A.1 (common-path Kronecker rank). If $h \neq 0$, then

$$\text{rank}(h \otimes A) = \text{rank}(A). \quad (8)$$

If $h = 0$, the rank is zero. Consequently, repeating a common ramp or other common scalar path over time does not separate mechanisms whose cross-moment loadings were already linearly dependent.

Proof. Regard h as a $T \times 1$ matrix. The rank identity for a Kronecker product gives

$$\text{rank}(h \otimes A) = \text{rank}(h) \text{rank}(A).$$

For nonzero h , $\text{rank}(h) = 1$; for $h = 0$, it is zero. Equivalently, if $Ac = 0$, then $(h \otimes A)c = h \otimes (Ac) = 0$. Conversely, when $h \neq 0$, select an index t with $h_t \neq 0$. The t -th q -row block of $(h \otimes A)c = 0$ is $h_t Ac = 0$, so $Ac = 0$. Thus the two designs have the same column null space and, for $h \neq 0$, the same rank. $\square$

Housing-specific interpretation. In a housing application, imposing a common temporal ramp on wealth, credit, supply, or demographic response signatures can add observations without adding an identifying direction. A longer common path may improve precision under a valid stochastic model, but it cannot by itself resolve mechanism collinearity. This conclusion concerns the imposed design and does not assert that the mechanisms truly share a common path in England.

A.4 Point identification and its dual certificate

Point identification here means uniqueness of β among all coefficient pairs satisfying equation (6). It is not a causal claim and does not address misspecification.

Proposition A.2 (focal identification against an unrestricted nuisance span). Assume that equation (6) is feasible. The focal coefficient β is unique for every feasible y if and only if

$$f \notin \text{col}(N). \quad (9)$$

Proof. Let (β_1, γ_1) and (β_2, γ_2) generate the same y . Their difference satisfies

$$f(\beta_1 - \beta_2) + N(\gamma_1 - \gamma_2) = 0. \quad (10)$$

If $f \notin \text{col}(N)$, equation (10) cannot hold with $\beta_1 - \beta_2 \neq 0$; hence $\beta_1 = \beta_2$.

Conversely, if $f \in \text{col}(N)$, choose a such that $f = Na$. From any feasible pair (β, γ) , the continuum

$$(\beta + t, \gamma - ta), \quad t \in \mathbb{R}, \quad (11)$$

generates the same y . Thus β is not unique. $\square$

The unrestricted-nuisance qualification is essential. Sign, box, shape, or cross-equation restrictions on γ can shrink the feasible values of β , possibly to a singleton, even when $f \in \text{col}(N)$. Such a conclusion is identification by the stated restrictions and must be reported as such.

Proposition A.3 (dual discriminator equivalence). The following statements are equivalent:

1. $f \notin \text{col}(N)$;
2. there exists $d \in \mathbb{R}^m$ such that $d^\top N = 0$ and $d^\top f = 1$.

When they hold, $\beta = d^\top y$ in the exact model.

Proof. Let P_N be the Euclidean orthogonal projector onto $\text{col}(N)$, and define $r = (I - P_N)f$. If $f \notin \text{col}(N)$, then $r \neq 0$, $N^\top r = 0$, and

$$r^\top f = r^\top (P_N f + r) = r^\top r > 0.$$

Therefore $d = r/(r^\top r)$ has the required properties. Conversely, if $f = Na$ for some a , then every d satisfying $d^\top N = 0$ also satisfies $d^\top f = d^\top Na = 0$, so no unit-loading discriminator exists. Finally, multiplying equation (6) by $d^\top$ yields $d^\top y = \beta$. $\square$

The vector d is a constructive diagnostic: it identifies the contrast of observed moments that annihilates every admitted nuisance column. It is not a substantive validation of those columns. A discriminator can disappear when a new, credible rival response is added to N .

A.5 Observation maps and missing-coordinate loss

Let $y^* \in \mathbb{R}^r$ be a latent or more richly measured response,

$$y^* = f^* \beta + N^* \gamma, \tag{12}$$

and let the available observations be $y = My^*$ for a known linear map $M \in \mathbb{R}^{m \times r}$. The observed response columns are then Mf^* and MN^* .

Proposition A.4 (identification after an observation map). Under unrestricted nuisance coefficients, the focal coefficient is identified from the observed vector if and only if

$$Mf^* \notin \text{col}(MN^*). \tag{13}$$

Equivalently, observed identification fails if and only if there exists $a \in \mathbb{R}^k$ such that

$$f^* - N^* a \in \ker(M). \tag{14}$$

Moreover, $\text{rank}(M[f^* \ N^*]) \leq \text{rank}([f^* \ N^*])$: an observation map can preserve or destroy design rank, but cannot increase the rank of the latent design.

Proof. Apply Proposition A.2 to the observed equation

$$y = (Mf^*)\beta + (MN^*)\gamma.$$

Failure of condition (13) means that $Mf^* = MN^*a$ for some a , which is exactly $M(f^* - N^*a) = 0$, proving equation (14). The rank inequality is the standard inequality $\text{rank}(MD) \leq \text{rank}(D)$, applied to $D = [f^* \ N^*]$. $\square$

If M selects observed coordinates, $\ker(M)$ consists of vectors supported on missing coordinates. Proposition A.4 then makes the loss mechanism explicit: a focal-minus-rival contrast may be visible in the richer system but live entirely in coordinates absent from the public observation map. Observing more dates of the same mapped coordinates need not recover that contrast.

Housing-specific interpretation. Candidate missing bridges include a mapping from property transitions to household tenure, a mapping from ownership or letting states into measured prices, and a mapping from lifecycle groups into the published age categories. The proposition says only that such coordinates *could* distinguish mechanisms. It does not establish that a particular restricted-access linkage is accurate, licence-compatible, or sufficient, nor that it would distinguish wealth from every admitted rival.

A.6 Weighted least-squares profile geometry

For noisy or overidentified moments, let $W \in \mathbb{R}^{m \times m}$ be symmetric positive definite and consider

$$Q(\beta, \gamma) = (y - f\beta - N\gamma)^\top W (y - f\beta - N\gamma). \quad (15)$$

Let C be any nonsingular square root satisfying $C^\top C = W$, write $\tilde{y} = Cy$, $\tilde{f} = Cf$, and $\tilde{N} = CN$, and let $P_{\tilde{N}}$ be the Euclidean orthogonal projector onto $\text{col}(\tilde{N})$. Define

$$r_y = (I - P_{\tilde{N}})\tilde{y}, \quad r_f = (I - P_{\tilde{N}})\tilde{f}. \quad (16)$$

Proposition A.5 (profile objective and curvature). With γ unrestricted,

$$Q_p(\beta) := \inf_{\gamma} Q(\beta, \gamma) = \|r_y - \beta r_f\|_2^2. \quad (17)$$

If $r_f \neq 0$, then

$$\hat{\beta} = \frac{r_f^\top r_y}{r_f^\top r_f}, \quad (18)$$

$$Q_p(\beta) = Q_p(\hat{\beta}) + (r_f^\top r_f)(\beta - \hat{\beta})^2. \quad (19)$$

If $r_f = 0$, the profile is flat: $Q_p(\beta) = \|r_y\|_2^2$ for all β . Furthermore, $r_f = 0$ if and only if $f \in \text{col}(N)$.

Proof. For fixed β , minimizing equation (15) over γ is Euclidean least squares after premultiplication by C . Its residual is the projection of $\tilde{y} - \beta\tilde{f}$ onto the orthogonal complement of $\text{col}(\tilde{N})$, giving equation (17). Expanding the square and completing it in β gives equations (18) and (19) when $r_f \neq 0$. When $r_f = 0$, the term depending on β vanishes. Finally, because C is nonsingular, $Cf \in \text{col}(CN)$ if and only if $f \in \text{col}(N)$. $\square$

Thus $r_f^\top r_f$ is the profile curvature, or design information, for the focal coefficient against the admitted span. Near-zero curvature warns that apparently precise coefficient estimates may be dominated by scaling, regularisation, or restrictions. The formula is an algebraic property of the supplied f, N, W ; it is not evidence that W is the true sampling precision or that $\hat{\beta}$ is causal.

A.7 Sharp conditional focal sets

Now impose explicit bounds. Let

$$\Gamma = \{\gamma \in \mathbb{R}^k : \ell \leq \gamma \leq u\}, \quad B = [b_L, b_U], \quad (20)$$

where inequalities are componentwise and either focal endpoint may be infinite. The conditional feasible set and its focal projection are

$$\Theta(y) = \{(\beta, \gamma) : y = f\beta + N\gamma, \beta \in B, \gamma \in \Gamma\}, \quad (21)$$

$$\mathcal{I}_\beta(y) = \{\beta : \text{there exists } \gamma \text{ with } (\beta, \gamma) \in \Theta(y)\}. \quad (22)$$

Proposition A.6 (sharp set and linear-program endpoints). The set $\mathcal{I}_\beta(y)$ is an interval, a ray, the real line, a singleton, or the empty set. When nonempty, its lower and upper endpoints in the extended real line are the values of the linear programs

$$\underline{\beta} = \inf_{\beta, \gamma} \beta \quad \text{subject to } (\beta, \gamma) \in \Theta(y), \quad (23)$$

$$\overline{\beta} = \sup_{\beta, \gamma} \beta \quad \text{subject to } (\beta, \gamma) \in \Theta(y). \quad (24)$$

Every finite optimum is attained. These endpoints are sharp conditional on the equality, the columns, and the bounds: every value between two feasible finite endpoint values is generated by a feasible coefficient pair.

Proof. The set $\Theta(y)$ is a polyhedron because it is defined by linear equalities and inequalities. Its projection onto the scalar β -axis is also a polyhedron, hence has one of the listed forms. Linear objectives over a nonempty polyhedron attain any finite optimum, giving equations (23)–(24). If (β_L, γ_L) and (β_U, γ_U) are feasible endpoint witnesses, then for any $\lambda \in [0, 1]$,

$$\lambda(\beta_L, \gamma_L) + (1 - \lambda)(\beta_U, \gamma_U)$$

is feasible by convexity. Its focal coordinate covers every value between β_L and β_U , proving sharpness. $\square$

An empty set is not a failed optimisation to be silently repaired: it rejects the joint equality and restrictions. An unbounded endpoint is likewise a scientific result. Most importantly, mathematical sharpness is conditional sharpness. Boxes chosen after inspecting y , response columns calibrated from the focal column, or rival sets omitted for convenience do not become empirically credible merely because the endpoint programs are solved exactly.

A.8 Why stocks do not identify gross tenure flows

Let $x_{jt} \geq 0$ be the level of the stock in tenure state j at the start of period t , for $j = 1, \dots, K$. Let $F_{ij,t} \geq 0$ be the number of units transitioning internally from state i to state j , $i \neq j$, and let q_{jt}

collect net external additions to state j (entries less exits, new supply less removals, and any other declared boundary terms). The accounting identity is

$$d_{jt} := x_{j,t+1} - x_{jt} - q_{jt} = \sum_{i \neq j} F_{ij,t} - \sum_{k \neq j} F_{jk,t}. \quad (25)$$

When all internal states and external boundary terms are consistently measured, $\sum_j d_{jt} = 0$.

Proposition A.7 (gross-flow non-identification from stocks). Suppose $K \geq 2$, only the net changes d_t are observed, and no capacity, turnover, or upper-bound restriction is imposed on F_t . If one nonnegative flow matrix satisfies equation (25), infinitely many do. In particular, for any distinct a, b and any $c \geq 0$, adding c to both $F_{ab,t}$ and $F_{ba,t}$ leaves every d_{jt} unchanged. Hence neither total gross internal flow nor the two opposing flows of a pair are point-identified by stock changes alone.

Proof. The added a -to- b flow subtracts c from the balance of state a and adds c to the balance of state b . The added reverse flow does the opposite. The net effect is zero in every state, while total gross flow rises by $2c$. Since c is arbitrary, the observed stock changes admit a continuum of observationally equivalent gross flows. $\square$

Proposition A.8 (sharp lower bound on total internal flow). Assume that q_t is known, $\sum_j d_{jt} = 0$, and transitions are allowed between every pair of distinct states. Let

$$G_t = \sum_i \sum_{j \neq i} F_{ij,t} \quad (26)$$

count each internal transition once. Then

$$G_t \geq \sum_j (d_{jt})_+ = \frac{1}{2} \sum_j |d_{jt}|, \quad (27)$$

and the bound is sharp. State-specific gross inflow and outflow satisfy

$$\sum_{i \neq j} F_{ij,t} \geq (d_{jt})_+, \quad \sum_{k \neq j} F_{jk,t} \geq (-d_{jt})_+. \quad (28)$$

Proof. Write $I_j = \sum_{i \neq j} F_{ij,t}$ and $O_j = \sum_{k \neq j} F_{jk,t}$. Since $d_j = I_j - O_j$ and both terms are nonnegative, $I_j \geq (d_j)_+$ and $O_j \geq (-d_j)_+$. Summing the first inequality over j gives $G_t = \sum_j I_j \geq \sum_j (d_j)_+$. The zero-sum condition implies $\sum_j (d_j)_+ = \sum_j (-d_j)_+ = \frac{1}{2} \sum_j |d_j|$.

To attain the bound, treat every state with $d_j < 0$ as supplying $-d_j$ units and every state with $d_j > 0$ as demanding d_j units. Because total supply equals total demand and the transition graph is complete, a nonnegative transportation plan can send exactly the required amount from shrinking to growing states, with no counterflow or cycle. Its total flow is $\sum_j (d_j)_+$. $\square$

If the transition graph is restricted, equation (27) remains a lower bound when a feasible flow exists, but it need not be sharp. For two states in a closed system, $d_2 = F_{12} - F_{21}$, so $F_{12} \geq (d_2)_+$

and $F_{21} \geq (-d_2)_+$; even there, the two gross flows remain unbounded above without additional restrictions. With three or more states, a rise in one state and fall in another generally does not impose a positive lower bound on the *direct* flow between that particular pair, because the balance can be routed through other states.

Two further qualifications matter in housing data. First, if q_t is unobserved and unrestricted, internal flow need not have a positive lower bound: setting $q_t = x_{t+1} - x_t$ and $F_t = 0$ satisfies the accounting identity. If instead q_t belongs to a declared feasible set $\mathcal{Q}_t$, the best stock-based lower bound is

$$\inf_{q \in \mathcal{Q}_t: \mathbf{1}^\top(x_{t+1} - x_t - q) = 0} \frac{1}{2} \|x_{t+1} - x_t - q\|_1, \quad (29)$$

provided the associated internal-flow problem is feasible. Second, changes in tenure *shares* alone do not determine changes in tenure levels when the total number of households or dwellings changes. The level denominator and the household–dwelling measurement bridge must be specified before applying the accounting bounds.

Housing-specific interpretation. An owner-occupation stock decline and a private-rental stock increase are net accounting facts, not measurements of gross owner-to-landlord purchases, landlord-to-owner sales, or household moves. Applying equation (27) requires compatible levels, boundary terms, units, geography, and timing. It cannot attribute the residual flow to the wealth of buyers without an additional buyer-resource measurement and a credible behavioural design.

A.9 A cautious design value for an additional measurement

The preceding results can rank candidate measurements without claiming that a dataset has policy or welfare value. Return to the latent columns f^*, N^* . For an observation map M and positive-definite weight W , define the focal separation index

$$J(M, W) = \min_{a \in \mathbb{R}^k} \|W^{1/2} M(f^* - N^* a)\|_2^2. \quad (30)$$

This is the squared weighted distance from Mf^* to $\text{col}(MN^*)$. It is positive exactly when the focal column is identified against the admitted nuisance span under M .

Let (M_0, W_0) describe existing moments. A candidate additional measurement has map H and positive-definite weight W_H , with block weights used only as a design convention. Define

$$J_0 = \min_a \|W_0^{1/2} M_0(f^* - N^* a)\|_2^2, \quad (31)$$

$$J_1 = \min_a \left\{ \|W_0^{1/2} M_0(f^* - N^* a)\|_2^2 + \|W_H^{1/2} H(f^* - N^* a)\|_2^2 \right\}. \quad (32)$$

Proposition A.9 (nonnegative design value of an added coordinate). Under the fixed latent columns, fixed nuisance set, and block-weight convention above, $J_1 \geq J_0$. Equality holds if and only if at least one minimiser of the old problem also satisfies

$$H(f^* - N^* a) = 0. \quad (33)$$

In particular, if $J_0 = 0$, the additional measurement restores point identification against the admitted span if and only if there is no a satisfying both

$$M_0(f^* - N^*a) = 0 \quad \text{and} \quad H(f^* - N^*a) = 0. \quad (34)$$

Proof. For each a , the objective in equation (32) is the old nonnegative objective plus a nonnegative term. Taking minima therefore gives $J_1 \geq J_0$. Both are finite-dimensional positive-semidefinite quadratic problems and attain their minima. Equality requires a minimiser of the new problem to attain the old minimum while making the added nonnegative term zero; this is exactly condition (33). When $J_0 = 0$, the old minimisers are precisely the solutions to the first equality in equation (34). The stacked design has positive separation exactly when none of them also satisfies the second equality. $\square$

Proposition A.9 defines *design value*, not the expected social value of information. It assumes the old and new response columns are correctly specified on a common scale, the nuisance set remains fixed, and the new measurement does not introduce selection, linkage, or transport error. A practical acquisition decision must also account for coverage, sampling error, revision, linkage quality, licence constraints, privacy, cost, and the risk that a defensible enlargement of N^* absorbs the apparent new separation.

In the housing application, a candidate measurement is most informative for mechanism separation when it observes a coordinate on which every currently equivalent focal–rival contrast differs. Buyer–resource links, gross property status changes, or household–dwelling reconciliation may be candidates, but Proposition A.9 does not presume that any available dataset measures them well or that any one of them is sufficient. The proposition supports a value-of-information *design audit*; it does not support an empirical England contribution estimate.

A.10 Implications for the empirical claim ceiling

The results imply a hierarchy of defensible statements:

1. A full-rank maintained design establishes only that its maintained columns are algebraically distinct.
2. If a credible rival span contains the focal column, unrestricted unique attribution fails, even when the maintained design alone has full rank.
3. A nonzero dual discriminator certifies separation only against the currently admitted columns and observation map.
4. Box or shape restrictions may deliver a sharp conditional interval, but the interval is empirical only after the restrictions and response sets are independently sourced and transported.
5. Stock changes can support accounting lower bounds under declared boundary assumptions; they do not reveal gross flows or buyer wealth.

6. Additional data have positive mechanism-separation value only to the extent that they break the relevant observational equivalences after measurement error and an enlarged rival set are considered.

Accordingly, the formal results justify a pivot from unrestricted unique attribution toward partial identification and measurement design. They do not reverse a failed identification gate, establish that a particular mechanism caused England’s tenure changes, or authorise a numerical historical contribution bound without the separate empirical response-set package described in Section A.1.

References

- Albuquerque, D., Lazarowicz, T., and Lenney, J. (2025). Monetary transmission through the housing sector. Staff Working Paper 1115, Bank of England.
- Bank of England (2023). Has the private rental sector been shrinking? Bank Overground. Published 12 September 2023.
- Bank of England (2026). Quoted household interest rates: Mortgage series IUMBV34 and IUMZID4. Monthly quoted two-year fixed 75 percent LTV owner-occupier and buy-to-let mortgage rates; series published 7 July 2026.
- Bracke, P. (2021). How much do investors pay for houses? *Real Estate Economics*, 49(S1):41–73.
- Castellanos, J., Hannon, A., and Paz-Pardo, G. (2024). The aggregate and distributional implications of credit shocks on housing and rental markets. Working Paper Series 2977, European Central Bank.
- Cima, S. and Kopecky, J. (2026). Housing policy, homeownership, and inequality. *Journal of Housing Economics*, 71:102118.
- Drechsel-Grau, M. and Greimel, F. (2026). Falling behind: Has rising inequality fueled the american debt boom? *Review of Financial Studies*, 39(2):459–517.
- Gomez, M. and Gouin-Bonenfant, É. (2024). Wealth inequality in a low rate environment. *Econometrica*, 92(1):201–246.
- Greenwald, D. L. and Guren, A. (2025). Do credit conditions move house prices? *American Economic Review*, 115(10):3559–3596.
- Head, A., Lloyd-Ellis, H., and Stacey, D. (2023). Heterogeneity, frictional assignment, and homeownership. *International Economic Review*, 64(3):1265–1308.
- HM Land Registry (2026). UK house price index: Data downloads may 2026. Statistical data set published 22 July 2026.

- Livingston, M., Pannullo, F., Bowman, A. W., Scott, E. M., and Bailey, N. (2021). Exploiting new forms of data to study the private rented sector: Strengths and limitations of a database of rental listings. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 184(2):663–682.
- Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer Series in Statistics. Springer, New York, NY.
- Mian, A., Straub, L., and Sufi, A. (2021). Indebted demand. *Quarterly Journal of Economics*, 136(4):2243–2307.
- Ministry of Housing, Communities and Local Government (2025). Live tables on housing supply: Net additional dwellings, table 120. Components of housing supply and net additional dwellings, England; updated 20 November 2025.
- Ministry of Housing, Communities and Local Government (2026a). English housing survey data on tenure trends and cross tenure: FA1201 (S106), age of household reference person by tenure. Live table containing 2024–25 data; updated 9 July 2026.
- Ministry of Housing, Communities and Local Government (2026b). Live tables on dwelling stock (including vacants): Table 104, by tenure, england (historical series). Table 104 updated 21 May 2026; landing page last updated 25 June 2026.
- Moktan, S. (2025). An equilibrium model of the markets for rental and owner-occupied housing. Job market paper, version dated 7 February 2025.
- Office for National Statistics (2022). Index of private housing rental prices. Time-series edition, version 25; released 19 October 2022.
- Office for National Statistics (2026). Price index of private rents, UK: Monthly price statistics. 22 July 2026 edition.
- Stevenson, G. (2019). The impact of inequality on asset prices when households care about wealth. MPhil thesis manuscript, University of Oxford. Unpublished manuscript dated May 2019; the URL is the official Oxford INET seminar record, not the thesis full text.
- Tamer, E. (2010). Partial identification in econometrics. *Annual Review of Economics*, 2:167–195.
- Zhang, F. (2015). Inequality and house prices. Job market paper, version dated 9 November 2015; revised 10 January 2016 and described by the author as under revision.